

Servlet, JSP & JDBC Interview Questions

Servlet Interview Questions

1. What is a Servlet?
2. What is the difference between Servlet and CGI?
3. Explain the lifecycle of a Servlet.
4. What are the different methods of the HttpServlet class?
5. What are the main steps to create a Servlet?
6. What are the advantages of using Servlets?
7. What is the difference between GenericServlet and HttpServlet?
8. How do you configure a Servlet in web.xml?
9. What is the web.xml deployment descriptor used for?
10. How do you map a Servlet to a URL pattern?
11. What are doGet() and doPost() methods?
12. What happens if both doGet() and doPost() are implemented in a Servlet?
13. When should we use doGet() and when doPost()?
14. What are Servlet attributes and how do you share data between Servlets?
15. What is RequestDispatcher? Explain forward() and include() methods.
16. How to redirect from one Servlet to another Servlet?
17. What is the difference between sendRedirect() and forward()?
18. How can you get initialization parameters from a Servlet?
19. What is the role of ServletContext and ServletConfig?
20. What is the difference between ServletConfig and ServletContext?
21. How do you manage session in Servlets?
22. What is the difference between URL Rewriting, Cookies, and HttpSession?
23. How does session tracking work internally?
24. How can you invalidate a session?
25. What are cookies? How do you create and read cookies in a Servlet?
26. How can you handle exceptions in Servlets?
27. What is the purpose of the Filter in Servlet?
28. What are Listeners in Servlet?
29. How can you restrict access to a Servlet?
30. What is the difference between Filter and Servlet?
31. What is the difference between SingleThreadModel and normal Servlets?
32. How to upload and download files using Servlets?
33. How do you handle multi-threading issues in Servlets?

- 34. What are the best practices for writing Servlets?
- 35. How to connect a Servlet to a database using JDBC?

JSP Interview Questions

- 36. What is JSP and why is it used?
- 37. What are the lifecycle methods of JSP?
- 38. Difference between JSP and Servlet?
- 39. What are JSP directives? Name and explain types.
- 40. What is the use of the `<%@ page %>` directive?
- 41. What is the use of `<%@ include %>` and ?
- 42. What are JSP scripting elements?
- 43. Difference between `<%= %>`, `<%! %>`, and `<% %>` tags.
- 44. What is JSP implicit objects? List some of them.
- 45. What is the difference between request and session implicit objects?
- 46. What are JSP actions?
- 47. What is the use of `<%= %>`, `<%! %>`, and `<% %>` ?
- 48. What is expression language (EL) in JSP?
- 49. How do you access request parameters and attributes in JSP?
- 50. What is JSTL? Why do we use it?
- 51. What are JSTL core tags?
- 52. Difference between `<%= %>` and `<%! %>` ?
- 53. How do you handle exceptions in JSP?
- 54. What is the purpose of `errorPage` and `isErrorPage` attributes?
- 55. How to perform pagination in JSP using JSTL?
- 56. How do you include dynamic content in JSP?
- 57. What are custom tags in JSP?
- 58. What is the difference between static and dynamic include?
- 59. What are the different scopes of JSP objects? (page, request, session, application)
- 60. How can you access Servlet variables in JSP and vice versa?
- 61. How do you prevent scriptlet code in JSP pages?
- 62. What are best practices for writing JSP pages?

JDBC Questions (with Servlet/JSP Integration)

- 63. What is JDBC?
- 64. What are the steps to connect Java with a database using JDBC?
- 65. What are the types of JDBC drivers?
- 66. Which JDBC driver is the fastest?
- 67. What is DriverManager?
- 68. What are Connection, Statement, and ResultSet objects?
- 69. Difference between Statement and PreparedStatement.

70. What are the advantages of PreparedStatement?
71. What is CallableStatement?
72. How to execute stored procedures using JDBC?
73. How to prevent SQL Injection in JDBC?
74. How to perform batch updates in JDBC?
75. How do you handle database transactions in JDBC?
76. What is auto-commit mode in JDBC?
77. How do you roll back a transaction in JDBC?
78. How to handle database connection pooling in web applications?
79. What is DataSource in JDBC?
80. How can you display database data on a JSP page using a Servlet controller?
81. How to pass form data from JSP to Servlet and insert into a database?
82. How to handle exceptions and close JDBC resources properly?

Project-Based / Practical Questions

83. Explain the flow: JSP → Servlet → JDBC → Database → JSP.
84. How do you perform CRUD operations using JSP, Servlet, and JDBC?
85. How can you validate a login using JSP & Servlet with JDBC?
86. How do you display data from a database table in a JSP using JSTL ?
87. What is MVC architecture in JSP/Servlet application?
88. How to upload a file and store file name in DB using JSP/Servlet?
89. How do you handle form validations on JSP & Servlet side?
90. How do you maintain user session after login using Servlet and JSP?
91. How do you secure JSP pages from unauthorized access?
92. How do you show error messages dynamically on JSP?

Performance & Best Practices

93. Why should business logic not be written in JSP?
94. How to reduce database load in a web application?
95. Why should database connections be closed in finally block?
96. How to implement connection pooling for performance improvement?
97. How to separate UI, Business, and DB logic in JSP/Servlet architecture?
98. How to use MVC pattern in JSP-Servlet applications?
99. What are some common exceptions in Servlets/JDBC and how to handle them?
100. How do you deploy your JSP/Servlet + JDBC application in Tomcat?